Muchen Ju

Contact Information

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Research Interest

Discrete Math, Probability and Combinatorics, and their applications.

Education

Fudan University, Shanghai, China
B.Sc. with Honors in Math and Applied Math — Sept.2022-Jul.2026
Thesis: Majority dynamics on sparse stochastic block models
University of Pennsylvania, Philadelphia, U.S.A.
Math Ph.D. Student — August 2026-Present

Exchange Experience

UC San Diego
UCEAP Exchange Student — Fall 2024

Workshops & Conferences

PKU Algebraic Combinatorics Experience 2024 [Website] — July-August 2024

PCMI 2025 Undergraduate Summer School [Website] — July 2025

Publications

Arborescences of random covering graphs, with Junjie Ni, Kaixin Wang, Yihan Xiao, [FPSAC 2025], [arXiv]
Sharp inner product correlations for hypercube bijections, with Ijay Narang. Submitted. [arXiv]
Majority dynamics on assortative sparse stochastic block models, with Ioana Dumitriu, Haixiao Wang. Submitted. [arXiv]
Majority Dynamics on Resampled Sparse Erdős–Rényi Graphs: Gaussian Winner Selection and Pace to Unanimity, with Ioana Dumitriu, Haixiao Wang. Submitted. [arXiv]

Selected Awards & Scholarships

National Scholarship(top 0.2% nationwide) — 2022-2023
17th Chinese Mathematical Competition, First Prize(ranked 1st in Shanghai in preliminary round) — April 2025
3rd Al-Khorezmi International Mathematical Olympiad, Gold Medal[Website](ranked #9 out of 212, highest in Team Fudan) — May 2025

Programming Languages

MATLAB, Python, LaTeX